Do people pre-activate phonological information sequentially or holistically?

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People can predict various types of linguistic information of a predictable word, including its phonological form [1], but it remains unclear whether phonological representations are pre-activated sequentially or holistically. According to the production-based model [2], listeners primarily rely on their production system to pre-activate the phonological representation of predictable words. Under this mechanism, phonological representations should be activated sequentially, from the first syllable to the last, analogous to the sequence of activation in language production. By contrast, another mechanism that may jointly account for prediction is prediction-by-association, in which listeners' mental representations of the context activate associated representations through spreading activation [2]. In this case, phonological representations are expected to be pre-activated holistically, since spreading activation operates on the word's representation as an integrated unit [3]. The present study used the visual-world eye-tracking paradigm to investigate whether phonological representations are pre-activated sequentially or holistically in native Mandarin speakers.

In our study, 102 native Mandarin speakers listened to 72 sentences, each containing a highly predictable target word. A visual display presenting one critical object and three distractors appeared 1000 ms before the target word onset. The target word and the critical object's names overlapped either at the first syllable (Head) or the second syllable (Tail), with an Unrelated condition serving as a baseline (see Fig. 1). The target and the competitor shared tone in 25 of the items. Predictive effects were assessed by comparing fixations on the critical object between the Head and Unrelated conditions and between the Tail and Unrelated conditions using a GAMM-based approach [4], which was also used to detect and compare the onsets of these effects.

The Tail condition elicited significantly more fixations than the Unrelated condition, whereas the Head condition did not differ significantly from the Unrelated condition (Fig. 2), suggesting that listeners predicted phonological information. The divergence between the Tail condition and the Unrelated condition occurred well before the target word onset, whereas no significant divergence was observed between the Head and the Unrelated conditions (Fig. 3).

The Tail effect in the absence of a Head effect is inconsistent with strictly sequential or holistic accounts. One possibility is that word-final segments can be pre-activated without pre- activation of word-initial segments. Because comprehenders may preferentially generate representations associated with the semantic head rather than modifier [5]. In Chinese lexical construction, where compounds can be further distinguished as left-headed or right-headed [6], such a head-oriented bias may give rise to position-specific prediction patterns. In our stimulus set, however, head position was not evenly distributed: the majority of target words were right-headed (around 57%), whereas only a small proportion were left-headed (around 6%), with the remaining items lacking a clear head-direction classification. This imbalance may have constrained our ability to detect a reliable left-position Head effect. Alternatively, listeners may have sequentially pre-activated both syllables, but pre-activation of the first syllable may have decayed before the analysis time window.

小李乘坐公交没有刷卡，而是投了几颗**硬币**。

gloss translation: 小李 Xiao-Li/ 乘坐 take/ 公交 bus/ 没有 not$_{\text{-perfective}}$/ 刷 tap/卡 card/ 而是 but/ 投了 insert$_{\text{-perfective}}$/ 几颗 several$_{\text{-classifier}}$/ 硬币 coin

"*Xiao Li did not tap his card when taking the bus but instead inserted a few* ***coins****.*"

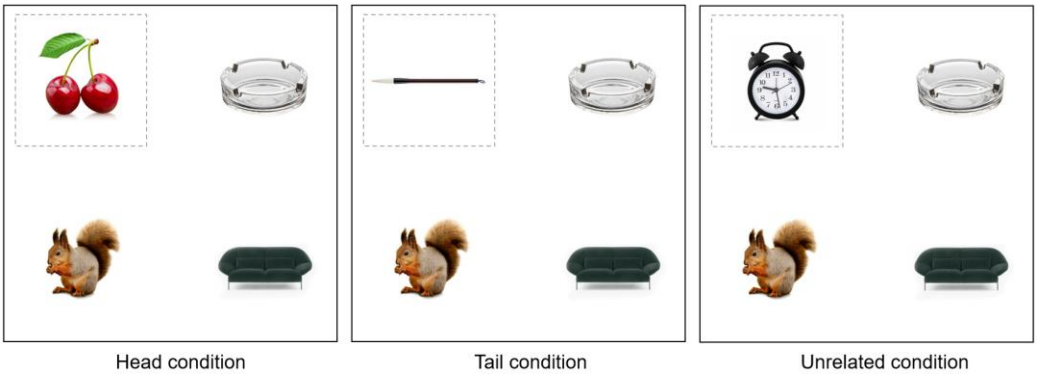

Figure 1 | An example of the visual stimuli for the sentence above. The bolded word at the end of the sentence is the target word (硬币/ying4bi4/ 'coin'). The objects in the dotted boxes (not shown to the participants) are the critical objects: a head competitor (樱桃/ying1tao2/ 'cherry') in the Head condition shares the first syllable with the target word. A tail competitor (毛笔/mao2bi3/ 'writing brush') in the Tail condition shares the second syllable with the target word. An unrelated object (闹钟/nao4zhong1/ 'clock') in the Unrelated condition is not phonologically related to the target. All the target and critical words are two-syllable words. All the words as well as distractors were selected to avoid orthographic similarity or semantic relatedness to the target or the context.

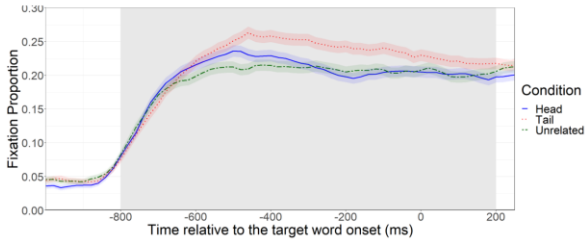

Figure 2 | Mean fixation proportion for critical objects in each condition with ±1 standard errors (the shaded area around each line). The grey shaded area indicates the analyzed time window. Time 0 indicates the onset of the target word.

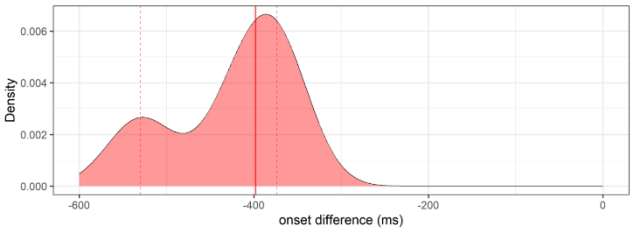

Figure 3 | Distribution of estimated temporal onset difference between the Tail and the Unrelated conditions. The vertical solid and dashed lines represent medians and 95% highest density intervals. The estimated onset was −398 ms [−530, −374].

References

[1] Ito (2024), *J. Mem. Lang*; [2] Pickering & Gambi (2018), *Psychol. Bull.*; [3] Magnuson et al., (2007), *Cogn. Sci.*; [4] Veríssimo & Lago (2025), *preprin*t. [5] Libben et al., (2003), *Brain Lang.*; [6] Packard (2000). Cambridge University Press